CHANGING BIOMES - HOW BIOMES HAVE BEEN ALTERED BY HUMAN ACTIVITIES

2014 Exam Question

Q18. Biomes are altered by human activity.
Discuss. **(80m)**

2012 Exam Question

Q18. Examine how any **three** of the activities listed below can impact on biomes:

- Early settlement and the clearing of forests
- The felling of tropical rain forests
- Intensive agricultural practices
- Industrial development.

(80m)

2011 Exam Question

Q18. The development of economic activities can alter biomes.
Discuss this statement with reference to appropriate examples that you have studied. **(80m)**

2009 Exam Question

Q 18. Assess how biomes have been altered by human activity. **(80m)**

A variety of human activities have changed biomes across the world. These activities include:

1. Deforestation
2. Settlement
3. Intensive agriculture
4. Industrial activity

The desert region of **California** is an example of an area where high population growth has affected the environment and as a result human activities have changed the desert biome.

1. Deforestation

Deforestation is the removal of a forest or trees where the land is thereafter converted to a non-forest use. Deforestation in California has changed the desert region.

In the 1800s The discovery of resources, including gold, led to a population explosion. This pattern of immigration to California still continues today. During the **gold rush** deforestation also began. Wood was plentiful in this region and lumber was the first building material to be produced industrially in northern California.

The **mining industry** also needed a great deal of wood, both to fuel the boilers at the mines and to build extensive canal systems. This demand for lumber helped create the logging industry **Saw mills** were therefore established throughout the region.

Improvements in transport, such as the railway, allowed the wood to be transported to rapidly growing urban centres such as San Francisco.

Trees were been rapidly cleared and new urban areas were springing up throughout the region. The result on the landscape was that once forested areas now consisted of scrub woodland or grassland.
California's best known trees are the **giant redwoods.** Since the gold rush over 96% of California's giant redwood forests have been removed. Structures built of redwood lasted longer than those built of pine so the wood was highly desired.
With less than 3% of the ancient redwoods still remaining in California, each remaining tree is precious. Logging and agriculture has wiped out nearly all of the 2,000,000 acres of redwood forest that once covered California
The majority of redwood forests are now under **federal ownership.** Felling of trees is prohibited in these forests. Preservation of forests has also saved the habitat of many plants and animals that were being endangered due to logging.

Settlement
Patterns of **human settlement** have had a huge effect on the environment in the desert area of California, including direct habitat destruction, contribution to greenhouse gases and obstruction of animal movement.
The **Europeans** arrived in North America in the 1700s. Prior to their arrival most of the native North Americans built their settlements to last from season to season.
With the arrival of the European came more permanent settlements. This increased further with the gold rush.
Continued population growth has led to the development of vast urban areas. Los Angeles is larger than County Dublin. Las Vegas has a population of 589,000 today and was only founded in 1905. By **2035,** California's population will approximately double to 64 million, if current population growth continues
Due to this continued development **over 500km^2** of natural habitat continues to be lost every year in California to residential or commercial use.
The result of this **urban growth** is that many species of plants and animals, including the desert tortoise are in danger of extinction. Illegal collection for the pet trade, destruction of their habitats by urban area expansion and off-road vehicles are some of the factors contributing to the desert tortoise's decline.
The **mountain yellow-legged frog** is believed to have been removed from approximately 99% of its' historic range, and is an endangered species. The current estimate of adult frogs is thought to be less than 200 individuals. These populations are vulnerable to extinction due to human activities such as recreational activities (e.g. hiking, camping, biking& fishing) and water removal for ski operations.

Intensive agriculture
It is acknowledged that unregulated **grazing** led to ecological damage that is still visible in California today. Cattle grazing in California increased dramatically during the gold rush. Intense grazing destroyed much of the native vegetation thus altering the desert biome.
Varieties of **fast growing grass** were introduced from Europe. They were not able to survive and there was a rapid increase in weeds such as the tumbleweed. Recently varieties of African grasses have been introduced into the Sonora Desert.
Many desert areas have now been given protected status and development is prohibited.
California is the main **food producing** state in the US. In areas like the Central Valley intensive farming is practised producing a wide variety of crops.

Due to the **arid climate** artificial methods of irrigation are employed. The natural environment has been tamed but this comes at a price.

- Due to **water shortages** the water table has dropped. This has caused evaporation of water and has led to salinisation and the subsequent loss of vegetation cover.
- Due to **food demands** several thousand hectares of non-agricultural land, mainly desert, are being converted to intensive farming each year. This results in the loss of local species and habitat.
- **Dam construction** has halted the natural flow of many rivers. The Colorado River now ends 60km from the ocean. 750,000 hectares of delta wetlands have become barren mudflats.
- **Run-off** from irrigated land carries pesticides and fertilisers leading to the growth of algae. The water suffers an oxygen loss and is unable to support fish life.

Industrial activity

Deserts are among the **harshest** habitats on Earth. And while it may seem to be hospitable to nothing, the desert is teeming with life. The hardiest and most adaptable of species are often found in desert habitats, but they need protection from threats such as encroaching development.

More than 100 **unique species** of plants, including the Joshua Tree, grow in the 25,000-square-mile Mojave Desert that stretches from California to Nevada, and from Arizona to Utah. Wildlife there includes owls, deer and endangered mountain lions, and the landscape features geological wonders like Death Valley.

Aggressive attempts to **industrialise** the desert are now threatening the Mojave, which already hosts several military bases, tourist resorts and gold mines.

Conservationists stopped plans for a nuclear waste dump in the desert but now dozens of investors want to build solar energy farms there - a 600-turbine wind farm is scheduled for completion in 2015.

The **desert tortoise** has been in decline for decades, and the effect of human activities in the deserts — including the development of casinos and shopping malls took its toll long before construction began on the solar power plant. The solar farms will industrialise the desert on an unparalleled scale.

There are currently **proposals** for more than one million acres of energy developments on public lands in the California deserts. While not all proposals will be approved, these industrial-sized facilities will have lasting impacts.

One **solution** is to develop renewable energy on land with low ecological value—mining-scarred lands or abandoned agricultural fields. By reusing these sites, it is possible to create sustainable energy sources and protect the desert ecosystems.